



YOUR TECHNOLOGY & INNOVATION PARTNER

Advanced AI Training - Session 1

Ethical AI

Overview

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Introduction

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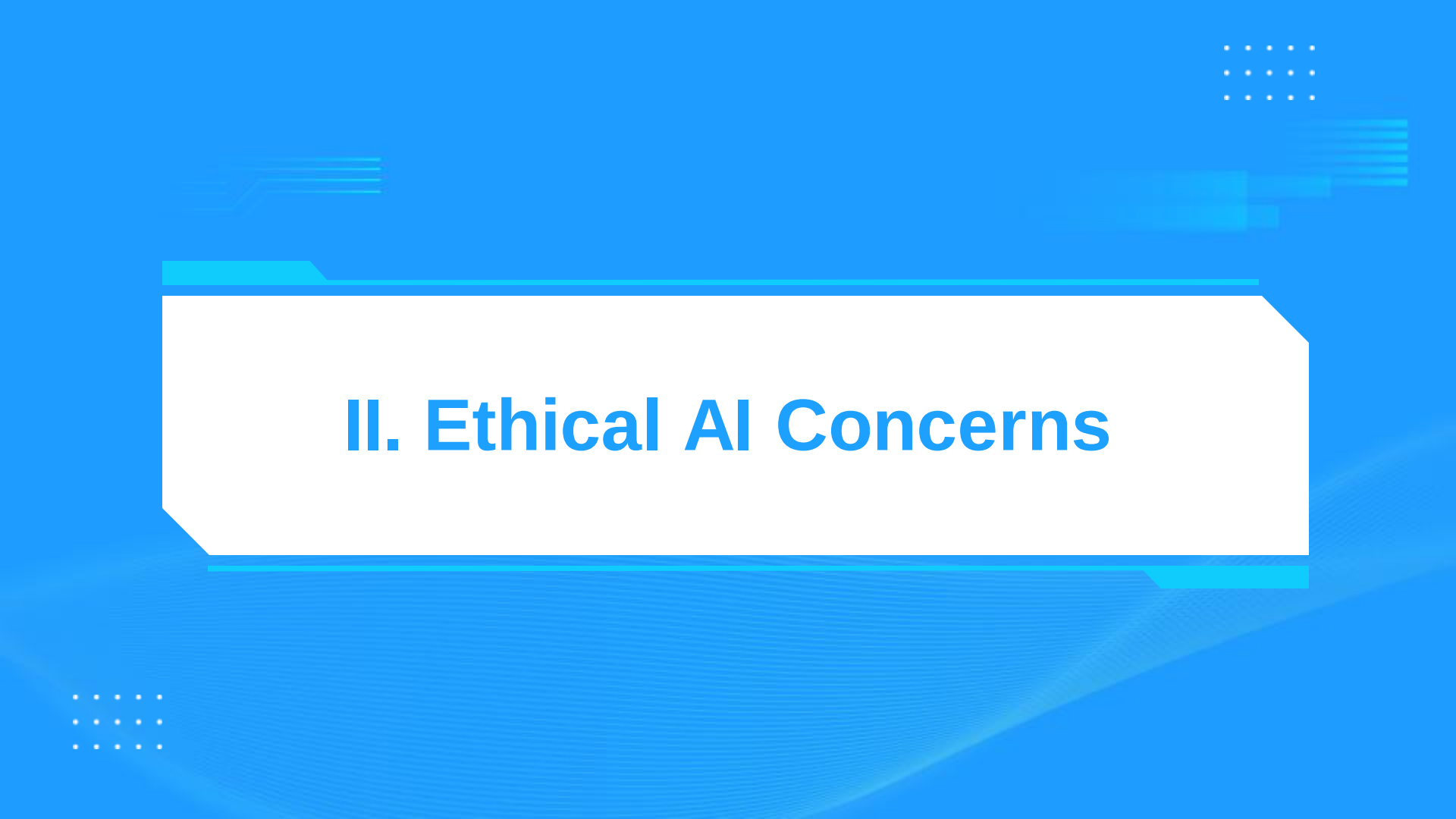
Key Takeaways



I. Introduction

Why ethical AI is necessary?





II. Ethical AI Concerns

II. Main Ethical AI Concerns and Their Impact

a) Bias and Discrimination

Inherent biases from historical data, leading to unfair or discriminatory



b) Transparency and Accountability

Requires large amounts of personal data, raising concerns about



c) Privacy and Security

Operate as "black boxes," making it difficult to understand or explain



a) Bias and Discrimination

- **Human vs Machine Bias:** Human bias is emotional and historical; machine bias comes from data and algorithms.
- **Algorithms Gain Bias from Context:** Math becomes ethical when used in real-world decisions.
- **Simpson's Paradox:** A gender bias case where department-level and university-level data gave conflicting results. Shows that context matters in interpreting bias.

What Is Bias?

Two Definitions of Bias:

Ethical/Cultural: Prejudice for or against individuals or groups.

Mathematical: Systematic error from incorrect data assumptions.

Humans Have Bias — So Do Machines :

Over 50 known cognitive biases in humans (e.g. bandwagon, placebo).

Machines reflect biases from data, design, or decisions.

Legal Concepts of Bias:

Protected Classes: Race, gender, religion—must not be treated unfairly.

Sensitive Characteristics: Features like race may be necessary for fairness (e.g. medical models).

Disparate Treatment: Unequal treatment due to protected class.

Disparate Outcome: Biased results regardless of intent.

Fairness Goals: Equal opportunity, demographic parity, predictive parity.

Data Bias Types and Examples

Types of Data Bias:

Sampling Bias: Data not representative (e.g. only data from Chicago).

Reporting Bias: Only certain outcomes reported (e.g. only violent police encounters).

Social Desirability Bias: Survey answers based on perceived acceptability.

Historical Bias: Feedback loops from past inequities (e.g. predictive policing).

Algorithmic Bias: Systemic issues from algorithm design.

Survivorship Bias: Focusing only on successful outcomes.

Key Insight:

Bias can exist at many stages: data collection, feature selection, model training, and evaluation.

Bias, Fairness & Trade-offs in AI Decisions

Loan Application Case Study:

Different groups (e.g., blue vs orange) have different risk distributions.

Four fairness strategies:

Max Profit: Highest gains but unequal treatment.

Group Unaware: Same model for all—may ignore real differences.

Demographic Parity: Same approval rate for all groups—may deny qualified applicants.

Equal Opportunity: Equal true positive rate—hard to achieve with overlap.

Key Takeaways:

- No single “fair” solution works in all cases.
- Bias and fairness require contextual choices and ethical trade-offs.
- AI does not invent bias—it reveals it, allowing for measurable and actionable improvements.

A Framework for Managing AI Bias

A – Awareness

Identify and measure all types of bias (e.g., accuracy, treatment, opportunity).

Focus on how outcomes affect protected classes.

D – Diversity

Use multiple metrics and perspectives.

Invite diverse viewpoints in analysis to detect blind spots.

A – Analysis

Ask why the bias exists—what causes it?

Examine unseen variables (like Mo's daycare responsibility).

Turn insights into real policy or model changes.

A Framework for Managing AI Bias

P – Publishing

Share bias findings transparently.

Be open to feedback and dialogue—bias management is collaborative.

T – Transform

Make ongoing improvements based on results and criterias.

1. Bias in Data Collection

Sampling Bias: Data doesn't represent population (e.g., celebrity images for facial recognition = underrepresentation of minorities/elders).

Survivorship Bias: Only successful cases are analyzed (e.g., WWII planes with visible damage vs. those that didn't return).

Collection Bias: Skewed by how data is gathered (e.g., online-only surveys miss offline voices).

2. Bias in Reporting

Reporting Bias: Only certain outcomes shared (e.g., only significant results).

Social Desirability Bias: Respondents answer to please or avoid judgment.

Publication Bias: Only positive or significant studies get published.

3. Bias in Historical & Algorithmic Systems

Historical Bias: Old patterns (e.g., biased hiring) embedded into new models.

Algorithmic Bias: Model decisions skewed by uneven or poor-quality data (e.g., health AI underpredicts complications for diverse patients).

Bias in Homogeneous vs Heterogeneous Data

Homogeneous Data Bias

Sample lacks diversity.

Leads to overfitting: good performance on sample, poor generalization.

Reinforces confirmation bias—assumptions go unchallenged.

Heterogeneous Data Bias

Differences between subgroups can introduce:

Measurement Bias: One group's data is collected differently.

Interaction Effects: Variables behave differently together (e.g., dimmer switch vs. circuit breaker).

Confounding Variables: Variables co-occur—hard to isolate effects.

 **Key Idea:**

Bias is not always bad—but must be measured, understood, and contextualized before deciding how to act on it.

Mitigating Data Bias in AI

Reduce Data Bias:

Use random, representative sampling.

Collect data from diverse sources & groups.

Apply validated, tested instruments (avoid DIY surveys).

Train data collectors to reduce unintentional bias.

Keep all variables—even sensitive ones—for testing bias, not always for modeling.

Ensure transparency, audits, and external reviews.

Involve diverse perspectives in data interpretation and model building.

Final Takeaway:

Good data $\neq$ bias-free data. The goal is to understand where bias exists, why it exists, and how it impacts your AI system before deciding what action is ethical, legal, or effective.

b) Transparency and Accountability

Trade-Off:

More complex models (e.g. deep learning) → better accuracy, less interpretability

Simpler models → clearer decisions, but may lack precision

Aspect	Transparency	Explainability	Aspect
Definition	Openness about data, models, training	Ability to understand decisions	Definition
Example - Tree	✗ (data hidden)	☑ (decisions are clear)	Example - Tree

NIST Principles for Explainable AI

1. Explanation – Every decision has a reason.
2. Meaningful – Explanations must be user-friendly.
3. Accuracy – Must reflect true system operations.
4. Knowledge Limits – Model should know its boundaries.

Use Case: In government or critical applications, explanations must be formal, consistent, and fair.

LIME Algorithm

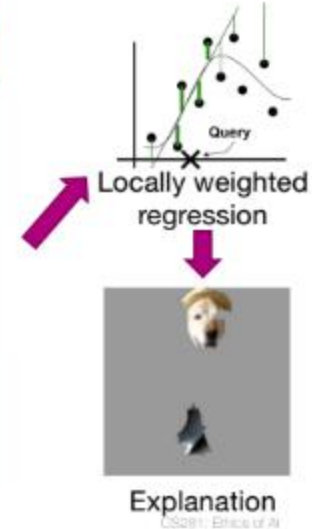
1. Take a single instance for which you want an explanation.
2. Generate synthetic data by perturbing the original instance.
3. Get model predictions on the perturbed data.
4. Fit a simple, interpretable model (e.g., a linear model) on the perturbed data and their corresponding predictions.
5. Use this simple model to explain which features contributed the most to the original prediction.

LIME Algorithm

Sampling example - images



Perturbed Instances	$P(\text{Labrador})$
	0.92
	0.001
	0.34



Explainability in Practice – Glaucoma Detection Example

- Problem: CNN detects glaucoma from eye scans – highly accurate, not interpretable.
- Solution: Use LIME (Locally Interpretable Models) to highlight which parts of the image influenced the model.
- Result: Helps clinicians verify model logic and potentially discover new diagnostic features.

Improves clinical trust, supports early diagnosis, and enables collaborative AI-human decision-making.

c) Privacy and Security

Why Privacy Matters: AI systems handle sensitive personal data.

Common Concerns:

- Who has access to your data?
- How is it being used?
- Can you control or delete it?

Rank Your Privacy Fears:

1. Government
2. Police
3. Big Tech (Google, Meta, etc.)
4. Public/Society
5. Employers or Universities

"Think critically about who controls your data—and why it matters."

Global Landscape of Privacy Laws

Privacy regulations are expanding worldwide

Data residency laws require local data storage

Varying approaches: U.S., EU, China

Proactive policy is key

Overregulation = inefficiency (e.g. airport security)

Underregulation = misuse and harm

"We must shape privacy laws with technical insight—not fear."

GDPR – European Union's Data Regulation

User Rights Under GDPR:

Access, Rectification, Erasure ("Right to be Forgotten")

Restrict Processing, Object to Use, Data Portability

Rights against Automated Decision-Making

Business Obligations:

Data protection by design & default

Impact assessments & breach reporting (within 72 hrs)

Data Protection Officer (DPO) required

Criticism:

High compliance costs → favors large corporations

May hinder innovation and AI advancement

Penalties: up to €20M or 4% of global revenue

California & U.S. State-Level Privacy

California Online Privacy Protection Act

- Requires posted privacy policies

- Defines how data is used, shared, and corrected

- 30-day cure period for violations

U.S. Approach:

- No broad federal law yet

- States take the lead (California, Virginia, etc.)

- Many companies adopt California rules nationwide

"U.S. privacy is patchwork—flexible but fragmented."

Final Reflections on AI & Privacy

Privacy regulations reflect cultural and political values

Balance needed between:

- Innovation & Regulation
- Security & Civil Rights
- Transparency & Control

Key Questions for Responsible AI:

- Who owns the data?
- Who benefits from AI systems?
- How are rights and power distributed?

"Good privacy policy isn't anti-technology—it makes technology trustworthy."

Case study

Address Resolution & AI in COVID Test Kit Deliver

The Scenario:

In 2021, the U.S. government announced mass mailing of COVID test kits to every household.

USPS had no prior coordination—had to respond quickly.

A Hopkins-affiliated team preemptively built an address verification system, saving an estimated \$4B in duplicate requests.

Case study

Address Resolution & AI in COVID Test Kit Deliver

The Problem: Incomplete or Unclear Address Resolution

Despite success, ~300,000 households couldn't receive kits due to address issues.

Problems included:

- Business addresses
- Shared/multi-family homes
- Duplicate or multiple listings
- Dormitories or informal residences

Affected groups were primarily low-income households.

Case study

Address Resolution & AI in COVID Test Kit Deliver

Initial ML Solution – Classification Model

Trained to classify requests into 5 categories (business, shared, multiple, duplicate, other).

Rather than relying solely on classification, they designed a two-tier AI + rules approach:

- Auto-approve “business” & “shared” addresses (sending a kit is cheaper than resolving)
- Use regular expressions to resolve “multiple” & “duplicate” addresses
- Send “other” cases for manual review



III. Applications in TMA

Solutions for Ethical AI in TMA

Prioritize Data Quality Over Quantity

Build Ethical Use Cases Around Human Impact

Enhancing Transparency and Accountability

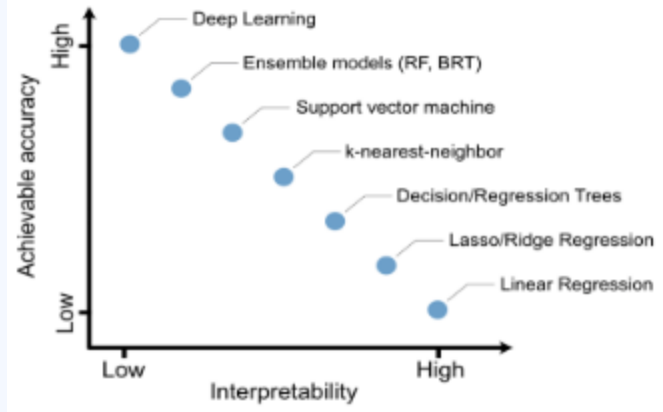
Protecting Privacy and Security

a) Prioritize Data Quality Over Quantity

- More data $\neq$ better results if the data is noisy or inconsistent.
- Invest early in annotation consistency and inter-annotator agreement.
- Poor labeling leads to technical debt that's expensive—or impossible—to fix later.

b) Enhancing Transparency and Accountability

- Develop Explainable AI (XAI) models to provide clear justifications for decisions.
- Maintain comprehensive documentation and audit trails.



c) Protecting Privacy and Security

Apply encryption and strict access controls to secure sensitive data.

Ensure transparent data handling policies and obtain informed consent.

Comply with regulations like GDPR and CCPA.

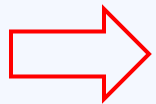


- Safeguarding Outputs
- Misinformation and Impersonation
- Spread misinformation
- Verification



Safeguarding Outputs

- Privacy
- Copyright and fair use
- Guardrails: LLMs can generate content that is considered (Violence and hate, Sexual content, Guns and illegal weapons, Regulated or controlled substances, Suicide and self-harm, Criminal planning).

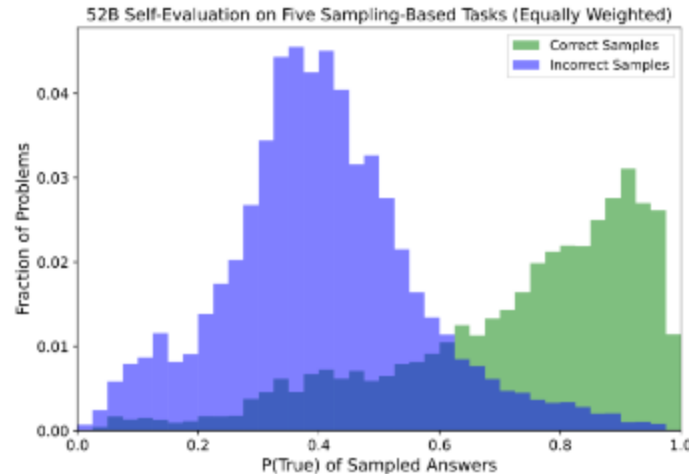


Apply default prompt of LLM

Verification

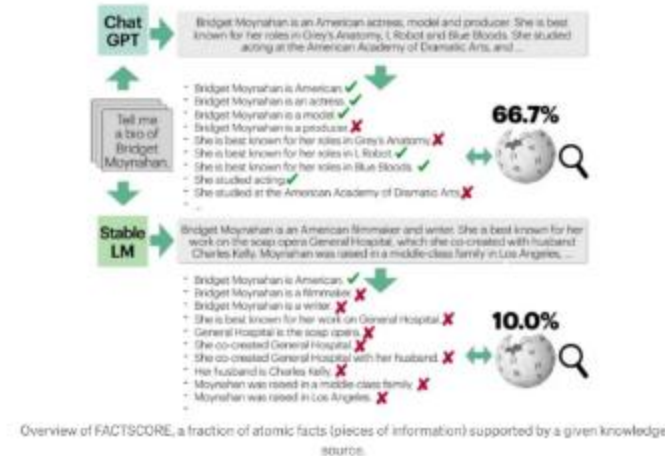
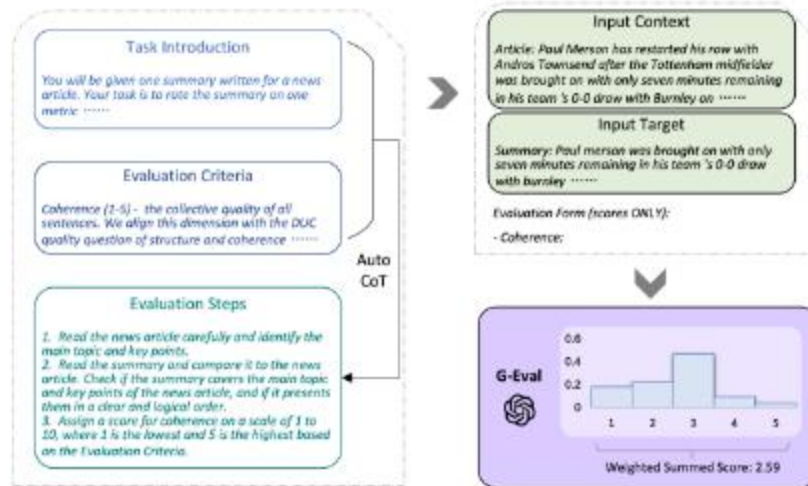
Research in 2022 when asking GPT

Measure model probability [Kadavath et al. 2022]



Resolve

- Evaluated for coherence, consistency, fluency and relevance
- Use external sources to verify results



d) Build Ethical Use Cases Around Human Impact

- Designing a Human-in-the-Loop (HITL) system requires carefully balancing human oversight with AI autonomy.
- Use AI where it directly improves outcomes for users—especially underserved ones.
- AI shouldn't replace decisions—it should enhance workflows.

- ## Modes of Human-AI Collaboration





IV. AI Ethics In Practice

IV. AI ethics in practice

FICO (Fair Isaac Corporation). Winner in the 2025 BIG Innovation Awards

- Comply with regulations like GDPR,CCPA ISO standards like ISO/IEC 23894
- Collect data across various industries
- Tracking of machine learning models across its entire lifecycle



V. Key Takeaways

Key Takeaways



Bias



Transparency



Privacy

Bias

Bias refers to systematic errors or unfair outcomes in AI that arise from data, algorithms, or human decisions

Key Sources:

- Data Bias: Historical inequalities, poor sampling, or underrepresentation.
- Annotation Bias: Inconsistent or unclear labeling standards.
- Algorithmic Bias: Models that favor one group over another due to skewed inputs or features.

Risks:

- Reinforces discrimination (e.g., predictive policing, health disparities)
- Harms vulnerable groups when bias goes unchecked
- Creates feedback loops that deepen inequality

Solutions:

- Use high-quality, diverse training data
- Monitor inter-annotator agreement and label consistency
- Evaluate models for equal accuracy across groups (not just global performance)
- Apply fairness-aware design from the start

Transparency in AI

Transparency is about being open and clear about how AI systems work—including the data, algorithms, decisions, and limitations.

Tools & Approaches:

LIME / SHAP: Highlight feature contributions for predictions

Decision trees / graphs: Help visualize model logic

NIST Framework: Explanation, Meaningful, Accuracy, Knowledge Limits

Challenges:

Trade-off between accuracy and interpretability

Humans often aren't transparent either—AI should be better, not perfect

Privacy concerns may limit full transparency

Privacy in AI

Privacy involves protecting personal and sensitive information from unauthorized access or misuse by AI systems.

Global Approaches:

- GDPR (EU): Strong data rights, but costly to implement
- California Privacy Law: State-level policies with specific compliance rules

Privacy Tensions:

- Trade-off between privacy and performance
- Excessive regulation can hinder innovation
- Delayed deployment (due to fear or compliance) can harm the people most in need

Best Practices:

- Minimize data collection
- Apply data anonymization & encryption
- Design for data protection by default
- Be clear about how data is used and shared

AI is powerful, but not **almighty**?



THANK YOU!